Case Study:

Part 1 – RAG-Based Question Answering over O2 Store Data

Introduction

The goal is to process the current sources about O2 stores and design an approach for enriching the LLM's knowledge with this information, so that the LLM can answer questions based on these sources.
A typical approach to enriching the knowledge of already pre-trained LLMs in this way is called RAG, which stands for retrieval-augmented generation. As part of this case study, you should therefore become familiar with this approach.
The goal is not necessarily to implement an enterprise-ready application, but rather to demonstrate that you understand the topic by building a working prototype and showing how such a solution could be implemented.

Data

Information about O2 stores can be downloaded in JSON format from the following URL:
o2.cz/servis/prodejny-export/export.json

Methodology

Take the source data and transform it into text chunks that contain relevant information for each O2 store, such as email address, phone number or opening hours.
Design an approach, or try to implement a method, that enriches the prompt of an LLM, such as ChatGPT, so that it can correctly answer questions like:

- What is the phone number of the O2 store in OC Metropole Zličín?
- What are the opening hours of the O2 store in OC Tesco, Hrádecká 1310, Sušice?

Expected output

A short presentation describing your approach and results.
Demonstration of working prototype.
Shared Python scripts or a Jupyter notebook, for example via GitHub.

Part 2 – Kubernetes

Create a Kubernetes cluster with at least two nodes. Deploy an application (can be a simple hello world) with components running on multiple nodes (see Figure 1). Parts of the application shall communicate with each other. Provide a definition of the cluster and the deployed application (code + description).

- What are basic parts of the Kubernetes cluster?
- How the Kubernetes cluster differs from the RedHat OpenShift?
- How would be realized deployment of the application (from the repository to the cluster)?
- How would be realized a rolling update (e.g. process of updating parts of the app in different pods/nodes, see Figure 2)?

Note: If you are unable to create the Kubernetes cluster on your PC, please contact us.

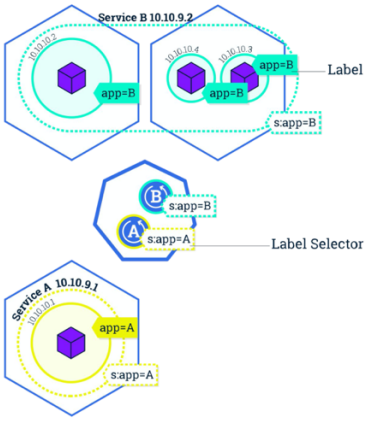


Figure 1: Multinode application

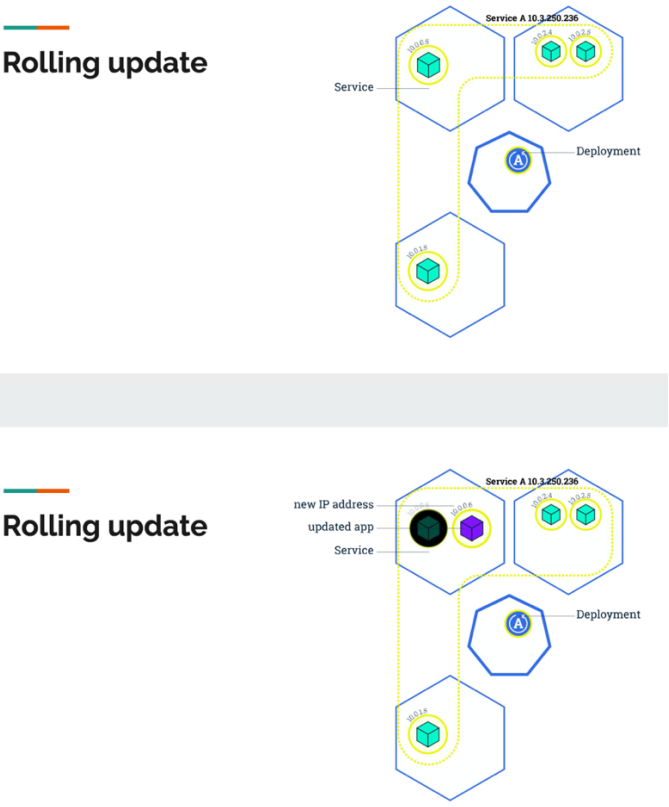


Figure 2: Rolling update

Timing

How much time do you have to prepare this? Again, it's up to you and your judgment of how much time is probably appropriate under these circumstances (but we appreciate efficient and simple solutions).

Asking and answering good questions is an inherent part of our job too. So, if you are going to ask, feel free to refer to brydl.ivo@o2.cz or filip.roskovec@o2.cz.

You have free hands in the selection of programing language or tools for completing the assignment, but keep in mind that our primary language is Python. We use PySpark for data processing, and our platform is in Azure Cloud. These tools are familiar to us, and we can quickly understand the solution. Show us how you can create a small piece of code that does the job in a nice and functional way, using your SW engineering skills.